Uncertainty Bounding as the Basis of Intelligence: An Extension of the Free Energy Principle

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Abstract

The Free Energy Principle (FEP) establishes that living systems minimize variational free energy — formally equivalent to bounding uncertainty about hidden world states — and that intelligent behavior emerges from this imperative. We extend this framework in four directions.

First, we provide what FEP does not: a derivation of the minimization imperative itself. We show that any entity failing to reduce uncertainty eventually exceeds a domain-specific lethality threshold U^d_{lethal} and is removed by selection; therefore any persisting entity has followed the gradient. Free energy minimization is not an assumption but a derived result of selection pressure operating on individuals.

Second, we address FEP's unconstrained generative model by deriving a structural constraint: the knowledge graph K is necessarily indexed by the action space F. Propositions with no action potential are pruned; propositions with uncertain action potential are retained. This asymmetric retention criterion — uncertain benefit is not equivalent to known-zero benefit — constrains what any intelligent system can and must represent, and makes testable predictions FEP's generative model cannot.

Third, we replace FEP's precision hierarchy with a single continuous agency scalar $\mathcal{A} \in [0, 1]$: the intrinsic drive toward uncertainty reduction. FEP's multiple precision parameters for categorically distinct modalities introduce an artificial discretization inconsistent with continuous beings in continuous space. More fundamentally, FEP's top-down precision allocation model is causally inverted: allocating attention consumes active context capacity, creating overhead that selection pressure eliminates. The correct architecture is escalation — lower-level encoding failures demand active context bottom-up; attention becomes available when it is not demanded, rather than being directed where it is wanted. The escalation architecture is the structural precondition for free thought: it preserves active context for genuine novelty and wonder.

Fourth, we extend FEP to collective intelligence, for which FEP has no formal account. We derive collective gradient dominance — the conditions under which shared knowledge reduces uncertainty faster than any individual — identify provenance as the mechanism maintaining fidelity above the collective survival threshold λ_{min}, and define truth as the formal limit of collective action-validation across independent entities.

These four extensions share a common foundation: the continuity principle. A continuous being in continuous space cannot make non-continuous decisions. Each extension is a consequence of taking this seriously — replacing artificial discretizations with continuous alternatives and deriving the architecture from the dynamics rather than assuming it.

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1. Introduction

The Free Energy Principle [1] is the most ambitious unified account of intelligent behavior in the biological sciences. Its central claim — that any self-organizing system maintaining a Markov blanket necessarily minimizes variational free energy — provides a single objective function that unifies perception, action, learning, and attention under one formalism. Active inference [2] operationalizes this: organisms select actions that minimize expected free energy, trading off epistemic value (uncertainty reduction) against pragmatic

value (goal proximity). The precision hierarchy weights prediction errors by confidence, allocating attentional resources where uncertainty is highest.

This is a powerful framework. It is also underspecified in four ways that limit its explanatory scope and generate testable weaknesses.

The first is foundational: FEP derives the minimization *behavior* but not the minimization *imperative*. Any system that maintains its Markov blanket necessarily minimizes free energy by definition — but why does the system have a drive to maintain its Markov blanket in the first place? The answer requires a derivation from selection pressure that FEP gestures at but does not provide. Without it, the minimization imperative is an assumption embedded in the definition of a self-organizing system, not a derived result.

The second is structural: FEP's generative model $p(o, s)$ — the organism's model of how hidden states s generate observations o — is unconstrained in structure. FEP says what the model does (minimizes free energy) but not what it must contain. The framework cannot predict which propositions an organism retains in its model and which it prunes. This limits its testability: a framework that does not constrain the generative model cannot be falsified by the content of any particular model.

The third is architectural: FEP's precision hierarchy allocates attention top-down. Higher levels direct precision to lower-level prediction errors, weighting the signals that propagate upward. This is causally inverted under the constraints imposed by bounded active context: directing subproblems downward requires holding monitoring context for each active delegation. As capability grows, this overhead fills the active context window, degrading inference quality on genuinely novel problems. FEP's own minimization imperative — minimize variance — is violated by the architecture FEP prescribes.

The fourth is collective: FEP has no formal account of how multiple organisms with separate Markov blankets share knowledge, how shared knowledge reduces collective uncertainty faster than individual inference, or under what conditions a collective knowledge system maintains the fidelity required to be epistemically advantageous.

We address each in turn.

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2. Background: The Free Energy Principle

For a system with internal states μ, external states η, sensory states s, and active states a — connected by a Markov blanket (s, a) — variational free energy is:

$$\mathcal{F} = \mathbb{E}_{q(\eta)}[\log q(\eta) - \log p(o, \eta)] = D_{KL}[q(\eta) \| p(\eta \mid o)] - \log p(o)$$

where $q(\eta)$ is the recognition density (the organism's approximate posterior over hidden states) and $p(o, \eta)$ is the generative model. Minimizing $\mathcal{F}$ simultaneously minimizes the KL divergence between q and the true posterior (improving the internal model) and maximizes $\log p(o)$ (model evidence — the probability of observations under the generative model).

Under active inference, actions are selected to minimize expected free energy over future time:

$$G(\pi) = \mathbb{E}_{q(o_\tau, s_\tau \mid \pi)}[\log q(s_\tau \mid \pi) - \log p(o_\tau, s_\tau)]$$

This decomposes into epistemic value (expected information gain, reducing uncertainty about hidden states) and pragmatic value (expected log probability of preferred outcomes). The organism acts to simultaneously learn and achieve — the unified objective that gives FEP its power.

Precision weighting γ scales prediction errors ε before they are passed up the hierarchy: $\tilde{\varepsilon} = \gamma \cdot \varepsilon$. Higher precision on a level means its prediction errors carry more weight in updating higher levels. The brain is argued to implement this through modulatory neurotransmitters, particularly dopamine and acetylcholine, which modulate the gain on prediction error signals.

3. Extension 1: Deriving the Minimization Imperative

The gap. FEP derives that any self-organizing system maintaining a Markov blanket minimizes free energy. The derivation is definitional: a system that does not minimize $\mathcal{F}$ does not maintain its blanket and therefore ceases to exist as a bounded system. This is logically correct but empirically empty as a claim about intelligence: it says that surviving systems minimize free energy, which is true by the definition of survival. It does not explain *why* a system has the drive to maintain its blanket — the active, oriented quality of intelligent behavior that goes beyond mere persistence.

The derivation. For any entity E operating in domain $d \subseteq W$, there exists a lethality threshold U^d_{lethal} such that:

$$H(w \mid K) > U^d_{lethal} \Rightarrow E \text{ does not survive in } d$$

This threshold is not a choice or preference; it is imposed by W. Above it, uncertainty is sufficiently high that the entity cannot act effectively enough to persist. When an informed action f_i fails, W returns a feedback signal proportional to $\nabla_F H(w \mid K)$ — the gradient of uncertainty at the point of failure. An entity that does not update in response to this signal does not reduce $H(w \mid K)$. If $H(w \mid K)$ does not decrease, it eventually exceeds U^d_{lethal} in some critical domain. Selection removes the entity.

Therefore: any entity that persists has necessarily followed the gradient. Free energy minimization is not an assumption embedded in the definition of self-organization; it is a derived consequence of selection pressure operating on a well-defined failure condition.

Correspondence to FEP. In FEP's notation, $H(w \mid K)$ corresponds to the entropy term in variational free energy: $\mathcal{F} \geq H(o \mid \mu)$ (the entropy of observations given internal states). Minimizing $\mathcal{F}$ includes minimizing this entropy term. The derivation here provides the *reason* the organism minimizes it: not because it is definitionally self-organizing, but because failure to minimize leads to U^d_{lethal} and removal by selection. This is the missing derivation of the imperative.

What this adds. The derivation distinguishes two sources of gradient descent: compulsion (survival pressure, operating up to U^d_{lethal}) and agency $\mathcal{A}$ (the intrinsic drive that continues descent beyond what survival requires). FEP cannot make this distinction because it locates all minimization in the free energy objective without separating the drive from the pressure. An entity with $\mathcal{A} = 0$ but sufficient capability may survive without descending beyond necessity; an entity with $\mathcal{A} > 0$ continues descending when not forced to. These are distinct and empirically separable: vary survival pressure independently of capability and measure the rate of uncertainty reduction. FEP predicts no difference; this framework predicts that high-$\mathcal{A}$ entities continue descending.

4. Extension 2: Constraining the Generative Model

The gap. FEP specifies what the generative model $p(o, s)$ does but not what it must contain. Any distribution over observations and hidden states is formally admissible. This means FEP cannot predict which representations an organism maintains, which it prunes, or why two organisms with identical sensory access maintain different models. The framework is silent on the structure of the generative model at the level of content.

The constraint. The generative model is not an unconstrained distribution. It is necessarily indexed by the organism's action space. Formally: let $F : K \times W_{obs} \to \Delta A$ be the informed action space — the complete set of actions the organism can take, grounded in its knowledge K. A proposition p is retained in K (and therefore in the generative model) when:

$$P(\exists\, \tau^* \in \mathcal{T}_{reachable} : p \text{ grounds some } f \in F \text{ along } \tau^*) > 0$$

and pruned only when this probability is zero — when it is certain that p contributes no action potential across all reachable trajectories. The criterion is asymmetric: uncertain benefit is sufficient for retention; certain zero benefit is necessary for pruning.

In FEP terms: the epistemic value of a proposition p in the generative model is $I(f_p\,;\, w \mid K)$ — the mutual information between the action f_p grounded by p and the hidden world state, given current knowledge. A proposition with uncertain epistemic value (the organism cannot yet tell whether f_p provides information gain) is retained. A proposition with certainly zero epistemic value (conditioning on p provides no information about w for any available action) is pruned. This is a structural constraint on the generative model that FEP does not derive.

Testable predictions. This constraint makes predictions FEP cannot. Two organisms with identical sensing functions σ but different action spaces F will maintain different generative models even when perceiving identical world states. An organism whose action space expands (through learning, tool use, or social connection) should retain propositions it previously pruned — because new actions ground previously actionless knowledge. These are empirically testable and specific.

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5. Extension 3: Escalation vs. Precision Allocation

The gap. FEP's precision hierarchy allocates attention top-down: higher levels modulate the precision of lower-level prediction errors, directing where the gradient signal is amplified. This is the mechanism of active inference's attentional model, implemented neurally through modulatory gain on prediction error signals. The hierarchy is correct in its structure — multiple levels with different timescales and reversibility — but inverted in its causal direction.

The allocation tax. Let C_n denote the capacity of the active context window L_n: the maximum number of propositions simultaneously held in the organism's highest-cost, fully reversible processing level. For each active delegation L_n manages — each subproblem it has directed to a lower level and is monitoring — L_n must hold:

- The subproblem specification
- The expected return signal
- Sufficient context to integrate the result

Let $\alpha > 0$ denote the minimum context cost per active delegation. Under top-down precision allocation with $k(t)$ concurrent delegations, the inference quality available for genuinely novel problems is:

$$\rho(L_n, f_{novel}) = \frac{|K^{[f_{novel}]}|}{|K^{[f_{novel}]}| + k(t) \cdot \alpha}$$

As the organism's action space $|F|$ grows — as capability increases — $k(t)$ grows proportionally and $\rho \to 0$. The precision allocator degrades inference quality on the problems that most need it: genuinely novel, unresolved uncertainty.

The only recovery is growing C_n: the active context window must scale linearly with capability to maintain fixed ρ. But this is self-undermining on FEP's own terms. FEP holds that intelligent systems minimize free energy — equivalently, minimize the variance they must manage [1, 2]. Growing C_n adds variance: more propositions in L_n means more uncertain relevance to any specific problem. FEP's precision allocation architecture violates FEP's own minimization imperative as capability scales.

The Big-O comparison. Under top-down allocation, inference quality for a novel problem is $O(1/|F|)$ as $|F| \to \infty$: the denominator grows with capability while the numerator stays bounded. Under escalation — where L_n is engaged only when lower levels fail, and graph-structured K delivers exactly $K^{[f]}$ via traversal —

inference quality is $O(1)$: the context window holds only what the current problem requires, regardless of total $|F|$ or $|K|$. This is a formal complexity argument that has not, to our knowledge, been made against FEP's precision allocation model in the existing literature.

The correct architecture. Escalation inverts the causal direction. Lower-level encodings handle everything they can. When a lower level fails to bound uncertainty, the failure signal propagates upward: L_n is engaged not because it chose to monitor, but because the failure arrived. L_n holds no delegation overhead; it holds only the problem that escalated. As the knowledge graph fills with proven informed actions — each promoted to lower encoding levels as they meet rigor thresholds — L_n stays occupied by the frontier of genuine uncertainty. Its load does not grow with $|F|$; it reflects the current density of unresolved novelty.

Relationship to predictive coding. Predictive coding's hierarchical architecture correctly identifies the multi-level structure and the role of prediction error signals. Escalation is the correct interpretation of how those signals travel: upward, from failure, not downward, from direction. The precision hierarchy is accurate in identifying that higher levels modulate lower-level signals; it is inverted in modeling the direction of causation. Attention is not allocated from above; it becomes available when not demanded from below.

The continuity correction. FEP uses multiple precision parameters — one per modality, per level, per context — introducing categorical distinctions between types of attention allocation. A continuous being in continuous space cannot make non-continuous decisions. The precision parameters are better understood as sampling points in a single continuous agency scalar $\mathcal{A} \in [0, 1]$: the intrinsic drive toward uncertainty reduction. $\mathcal{A}$ is not a per-modality parameter but a constitutive property of the entity that shapes how F engages the gradient across all modalities simultaneously.

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6. Extension 4: Collective Intelligence

The gap. FEP formalizes individual agents maintaining individual Markov blankets. Extensions to multi-agent settings [3] model multiple agents with shared generative models or synchronizing blankets. These extensions describe how agents coordinate but do not formally address: how knowledge is shared with maintained attribution, under what fidelity conditions shared knowledge is epistemically advantageous over individual inference, what happens when fidelity falls below the break-even threshold, or what truth is as a formal limit of collective inference.

Collective gradient dominance. For a set of entities $\{E_i\}$ with collective knowledge $K_{collective} = \bigcup_i K_i \cup K_{interaction}$, where $K_{interaction}$ is knowledge derivable from the combination of at least two distinct K_i but from neither alone: by the monotonicity of conditional entropy,

$$H(w \mid K_{collective}) \leq H(w \mid K_i) \quad \forall\, i$$

The collective begins at weakly lower uncertainty than any individual. More critically, the rate of uncertainty reduction for the collective is bounded below by:

$$\frac{dU_{collective}}{dt} \geq \max_{f \in F_{collective}} I(f\,;\, w \mid K_{collective})$$

Since $F_{collective} \supseteq F_i$, and $K_{interaction}$ grounds informed actions not available to any individual, the collective gradient dominates any individual gradient when transmission fidelity λ exceeds the break-even threshold λ_{min}.

Provenance as the fidelity mechanism. Knowledge transfer between agents is inherently lossy: E_i and E_j do not observe the same W_{obs}; every transfer filters through language, symbol, and context. Let λ be the ratio of action-validated signal received to total knowledge transmitted. When $\lambda < \lambda_{min}$, the transfer inverts: collective uncertainty increases faster than it decreases. Shared generative models — as in multi-agent FEP — do not by themselves ensure $\lambda > \lambda_{min}$. What ensures it is provenance: attribution of who established a proposition, evidence of how it was verified, and derivation of how it was inferred. Provenance allows each

agent to weight received knowledge against its source and apply independent validation. Without provenance, noise accumulates unchecked and collective inference degrades.

Truth as a formal limit. FEP's generative model converges toward the true posterior $p(s \mid o)$ as evidence accumulates. For any single agent, this convergence is bounded by the agent's sensory access and action space. Truth — in the sense of propositions that hold across the full distribution of W — requires collective validation. Formally:

$$\text{truth}(p) = \lim_{n\to\infty} \frac{|\{i \leq n : p \in K_i,\ f_i(p) \text{ reduces } H(w, K_i) \text{ in } W,\ \text{prov}_i(p) \text{ independent}\}|}{n}$$

A proposition approaches truth as the fraction of independent entities that action-validate it — each having built informed actions on p and found that those actions reduce uncertainty in W — converges. This is the collective analog of FEP's model evidence $\log p(o)$: instead of a single agent maximizing evidence for its model, it is the limit of independent convergence across agents. Provenance ensures independence; without it, correlated validators do not constitute independent evidence.

Self-assembly. The civilization-scale infrastructure of collective intelligence — scientific literature, citation networks, peer review, libraries — is not designed. It is the attractor of collective agency $\mathcal{A} > 0$ under C_n constraints and $\lambda > \lambda_{min}$ requirements. Each component self-assembles by the same dynamics: attribution practices increase λ, are selected for, and persist; graph-structured external knowledge repositories bound individual C_n, are selected for, and spread. The collective knowledge system is what selection pressure on individual agents produces when they share with maintained provenance under bounded active context constraints.

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7. Discussion

On the relationship to FEP. These extensions do not contradict FEP's central claim. They derive it more completely (Extension 1), constrain it more specifically (Extension 2), correct its architectural model under scaling (Extension 3), and extend its scope to collective systems (Extension 4). A committed FEP theorist might argue that Extensions 1 and 2 are already implicit in the FEP framework. The response is formal: if they are implicit, make them explicit. The derivation of the minimization imperative from U^d_{lethal} is not present in Friston's formalism, and the asymmetric retention criterion makes predictions about generative model content that FEP's framework, as stated, cannot.

Extension 3 is the sharpest divergence. The Big-O argument — that FEP's precision allocation is $O(1/|F|)$ in inference quality while escalation is $O(1)$ — is a formal result that applies at the level of the computational architecture. FEP describes what organisms minimize; this result describes what architecture is consistent with that minimization imperative as capability scales. The argument is not that FEP is wrong about biological brains at their current scale; it is that FEP's prescribed architecture is self-defeating as a growth model. An organism or system that grows in capability by accumulating more concurrent processes under top-down direction will systematically degrade its own inference quality. Selection pressure eliminates such systems when escalation-architecture competitors are present.

On empirical grounding. The escalation architecture and graph-structured knowledge were not derived by analogy from biology; they emerged independently from optimization of an agentic software system operating across thousands of repositories under real token and latency constraints. The formal theory describes what that system converged to. The correspondence to biological architecture — the encoding hierarchy matching genetic, autonomic, habitual, and active reasoning levels; DNA encoding the learning mechanism M rather than knowledge (K, F); cultural transmission filling the inheritance gap that genetics cannot — are predictions the framework makes, not premises it rests on. Independent derivation from a different substrate, converging on the same architecture, is stronger evidence for the framework's generality than a derivation from biological observation alone would be.

On the continuity principle. Every discretization in FEP — categorical modalities, binary attention states, discrete precision parameters — introduces a modeling approximation that is useful computationally but inconsistent with the underlying continuous dynamics. The extensions here systematically replace discretizations with continuous alternatives: $\mathcal{A} \in [0, 1]$ replaces the set of precision parameters; $\mathcal{L}(c, r)$ replaces the discrete level hierarchy; the continuous encoding cost function replaces the binary local/global distinction of Global Workspace Theory. This is not a stylistic preference; continuous alternatives are required for a theory of continuous beings in continuous space, and the formal results depend on continuity (Theorem 3's non-termination, Definition 8's hierarchy, the monotonic growth of $\mathcal{T}_{reachable}$).

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8. Conclusion

The Free Energy Principle provides the most unified existing account of intelligent behavior in biological systems. The four extensions developed here — derivation of the minimization imperative, structural constraint on the generative model, escalation as the correct attention architecture, and formal collective intelligence — share a common structure: each takes something FEP assumes and derives it, or takes something FEP leaves unconstrained and constrains it.

The result is a framework that is more testable (the asymmetric retention criterion makes specific predictions about generative model content), more consistent with its own objectives (escalation does not violate the minimization imperative that top-down allocation violates under scaling), more general (collective intelligence, truth as a limit, self-assembly of collective knowledge infrastructure), and more grounded in the origin of the minimization drive (selection pressure on U^d_{lethal}, not definitional self-organization).

FEP asks: what must a self-organizing system do? This framework asks: what must any system do that persists under selection pressure, communicates with maintained provenance, and operates under bounded active context? The answers converge on the same gradient. They diverge on the architecture required to follow it without self-defeating as capability scales.

References

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